Internet of Things: Informatic System for Metering with Communications MQTT over GPRS for Smart Meters

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***Abstract*—The measurement of consumption of public services such as drinking water and electric energy, and information management is one of the problems facing intelligent cities. In this article we describe the design of a computer system that allows the management and administration of consumption records made by smart meters that communicate through MQTT using mobile telecommunications networks. The results obtained allowed to determine the efficiency of the connectivity through MQTT over GPRS and to perform tests of the computer system in order to validate its functionalities.**

***Keywords*—MQTT,GPRS, Smart Meters, Mongo DB, JSON, M2M.**

I. Introduction

The Internet of Things has an important factor it is a integration of several technologies and communications solutions [1], one of the objectives of the IoT is develop a unique way of identification so that every object can be connected [2] principally by the wireless networks. But one of the challenging issues to support massive numbers of machine type communication (MTC) between them or a network node is to provide an efficient way for multiple access in the network and to minimize network overload as described in [3].In this respect, the current cellular systems has the cost-effectiveness, technological maturity and geographical coverage for gives support to the new form of communication is generally referred to as Machine-to-Machine (M2M) Communication, or Machine-Type Communication (MTC), while the involved devices are called Machine-Type Devices (MTD) [4].

Several research studies about IoT have been developed on wireless communications such as described in [4], these studies based on mobile cellular networks have considered the infrastructure deployed by mobile operators, the low cost that is expected to have the services based on MTD, considering a high volume of M2M market that allows economies of scale and, therefore, the interoperability of different technologies for small wireless networks such as IEEE 802.15.4 and Zigbee that is detailed in [5], [6], 6LowPAN [7], Bluethoot.

In the same way, technologies such as GSM, GPRS, EDGE and LTE becomes an attractive candidate to provide connectivity for smart cities. The 3G and 4G mobile network provides broadband bandwidth and supports to IoT mobility access environments, such as services monitoring, telemetry and health services [8]. In several countries the scalability of 3G network has to be supported by robust backhaul network. These networks implemented over several years are the perfect route to support the connectivity of Wireless Sensor Networks due to their coverage especially in places where it is not planned to deploy 4G networks. On the subject, in [9] described to conceptual design for implementation of a network sensor using arduino devices and multiplatform applications through OPC UA. In [10] design a of Wild Animals Monitoring System based on 3G and Internet of Things. In [11] propose the concept of adaptive message aggregation method for MQTT-SN protocol that optimizes power used by GPRS wireless connection during data transmission. The Precision Agriculture and Ecological Monitoring Through GPRS for sending data is described in [12].

On the other hand, the connectivity provided by the wireless technologies can be used several protocols to connect the user with the applications and the Internet, such as the protocol Message Queue Telemetry Transport

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(MQTT) [13], Constrained Application Protocol (COAP) [14], Extensible Messaging and Presence Protocol (XMPP) [15], Advanced Message Queuing Protocol (AMQP) [16], and WEBSOCKET [17]. In the same way, (MQTT) protocol is the most used for develop IoT projects as described in [18], the publish / subscribe protocols as MQTT are most suitable for end-applications that require massive access.

With respect to described, several research projects were developed of telemetry, focused in smart meters that used wireless networks. In [19]discusses various features, technologies and deployment of smart meters needs proper selection and implementation of a communication network of smart grid communication. This research [20] describe the development of digital water infrastructure using wireless sensor networks used MQTT and WiFi networks, likewise, in [21] developed a IoT system with MQTT over Wifi. Finally in [11] use MQTT and GPRS connectivity for telemetry applications.

Unlike previous works, there are works that study some protocols such as MQTT, REST, AMQP, over wireless technologies such as GPRS, LTE, WiFi, Bluethoot and design projects of IoT for smart meters only for obtain data of sensors with different protocols. However, in the studies mentioned above, the measurement of services such as drinking water and electric energy is not considered to require a two-way flow of information through the integration of advanced monitoring, communication and control technologies through a computer system that registers the Consumption of the service for pre-billing as well as control of smart meters. It is a latent problem for the providers of these services because they have not an informatics system for consume data management integrated with a intelligent meters of electric energy and water.

The paper has the following structure: Section II describes the general architecture of the system for intelligent meters, in Section III details develop and implementation MQTT over GPRS.In the section IV we describe the test and results. Finally in section V, we present the conclusions and future work.

II. General Architecture of the System for Data Management and Control

The General Architecture of the Management and Control system for intelligent drinking water and electricity meters has four layers as described in [22] and these are:

- Sensor layer: This layer is made up of smart meters of drinking water and electric energy, and is responsible for collecting the information for send to management and security layer through to network layer. We design and develop two real-time monitoring modules for both Smart Grid and Smart Water control systems with Arduino boards.
- Gateway and Network Layer: We consider that cellular communication (GSM Technologies) is the most ideal for this project because the MQTT is a protocol that needs low data bandwidth and GSM support the long-distance for IoT applications, likewise, the GSM networks have greater coverage than 4G networks, especially in rural areas and finally the motivation behind using this networks was the scalability factor.
- Management and Security layer: This layer manages the connection between intelligent devices and the application layer, using a MQTT broker. The security mechanisms of MQTT protocol are lightweight, this help to the constrained resources of the smart devices. For this project consider that the MQTT standard policies about authentication and authorization for the subscription of clients through message topics, are sufficient to provide safety to the sensor network.
- Smart Applications layer: In this layer employs intelligent computing technologies to save measurements information in a database and extract information for processing data for pre billing and provides a interface for users about to results of measurements of service consume and statistics. On the other hand, with the application developed We create knowledge about the consume history of a customer

As shown the General Architecture of the Management and Control system for intelligent drinking water and electricity meters above described in Fig. 1.

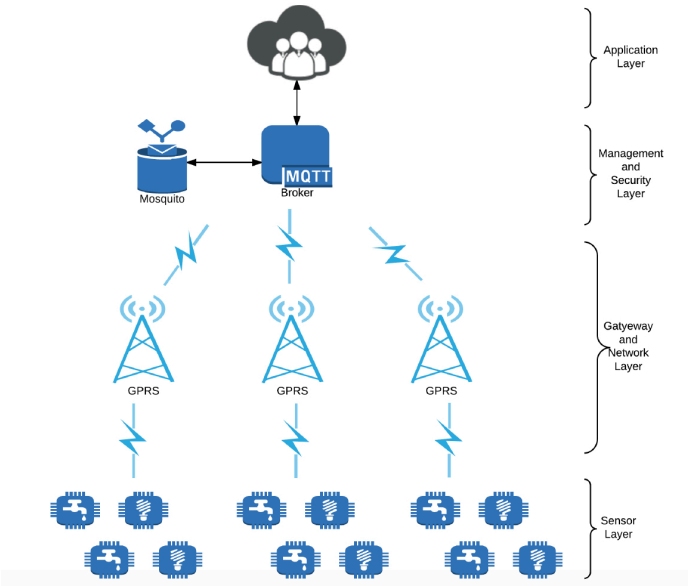


Figure 1: Architecture of the system

The most relevant elements that are part of the architecture are described below:

A. Management and Control System

The informatics system for management and control of the smart meters was implemented in a Apache Server [23] and developed in PHP [24]. This system was developed after a survey of use cases considering different user profiles and roles, considering the requirements and best practices in the management of users of Drinking Water and Electric Power Companies that described in [25].

As shown The general structure of the of informatics system of Management and Control of the services, in Fig.2

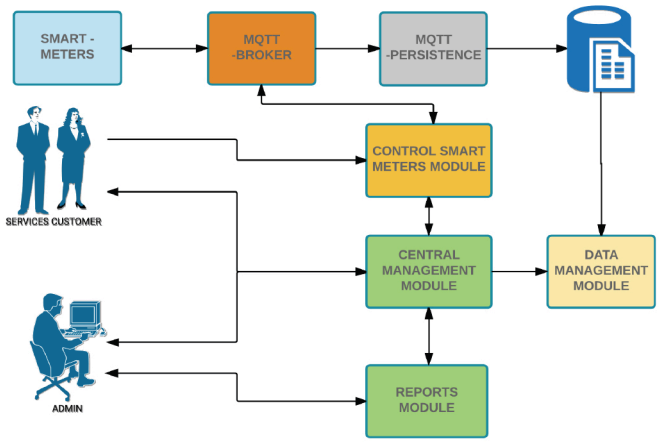


Figure 2: Informatics System Structure [26]

- Central Management Module: This module permits control and management different kind of users such as administrator and clients.
- Reports Module: This module allows to generate reports of daily, monthly or annual consumption of the two services or independently.
- Meter Control Module: Through this module, every smart meter are controlled for on/off, likewise allows control for groups of smart meters. And this module obtain by MQTT server the information such as, date, time, location for all smart meters, and power,voltage, current of the electricity meter and water flow of the water meter.
- Management Data Module: By means of this module the information about of services (electricity and water) that are sending by every smart meter are save in data base. The MongoDB [27] the database used for the development of this project. This data base save the data structure in JSON documents with a dynamic scheme. Mongo DB has characteristic such as: Ad hoc Queries, anyfield in a MongoDB document can be indexed, supports the type of primary-secondary replication, can be scaled horizontally using the concept of ”shard” [28].
- Persistence MQTT: For receive information of smart meters, we implemented to broker MQTT with that obtain the records of measurements made by smart meters. For this we configure a Client MQTT with Node JS this allows receive the messages for before connect to MongoDB and insert the smart meters data. Every smart meter have a ID, this help to send values of measurements concatenated with value of time accumulated and coordinates of location.

In general the informatics system allows obtain several statistics about costumers of services, data analysis, and develop reports per service user, group or sectors of users, per periods of analysis. Likewise, the most important target of this system is the obtain information on-line of the costumer of services for the pre-billing, this avoids problems in the manual registration or errors in the billing of basic services of drinking water and electricity.

B. Water Smart Meter

The Water Smart Meter was build with a Arduino Mega as the basis of this device, to which they connect a sensor of hall effect type YF-S201, module GPS type Neo-6m, Micro SD card memory, led display and shield GSM Type SIM900. This device accumulates the water flow for 15 minutes in Micro SD card and sends the information to the database using MQTT using the GPRS network. As shown in the structure of smart meter in Fig. 3.

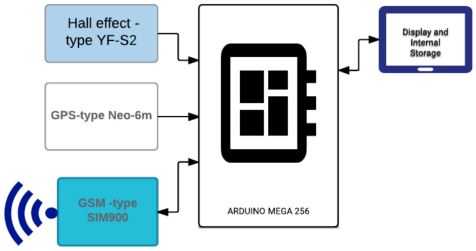


Figure 3: Water Smart Meter

C. Electricity Smart Meter

The Electricity Smart Meter was build with a Arduino Mega as the basis of this device, to which they connect a sensor CT (100A max clip-on current), module GPS type Neo-6m, shield emontx, shield GPS Type SIM900, Micro SD card memory and led display, this architecture is based in the reference [29]. This device send information about power (real and apparent), power factor, voltage, and sends the information to the database using MQTT using the GPRS network. The structure of smart meter as shown in Fig. 4.

III. Develop and Implementation MQTT over GPRS

We developed with MQTT because is a lightweight and simple messaging protocol [30] by its architecture of publish/subscribe, this allows the management thousands of remote clients, in our case smart meters of the drinking water and electricity. With the MQTT protocol the clients can subscribe to topics that pertain to them and thereby receive whatever messages are published to those topics [31]. The MQTT configuration for this project is described below:

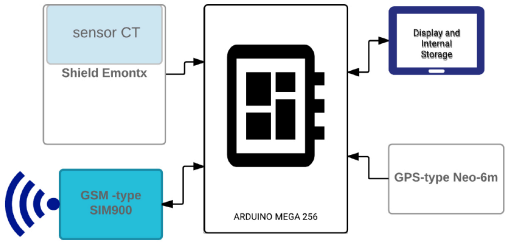


Figure 4: Electric Energy Smart Meter

- Topics and subscriptions: In [31] describes that MQTT are published to topics and clients sign up to receive particular messages by subscribing to a topic. In this regard, in this model a broker MQTT identify the status of a receiver which is smart meter before sending a command. A receiver also can identify the status of a broker before doing action.
- Publiblish/subcribe: MQTT subscriptions and publications can only be made on a specified set of topics, this protocol uses character strings to provide support of hierarchical topics [31].In this model, we configured such as clients the publisher and subscriber and called like theme to the message destination. Publishers send a specific topic of messages to message broker, subscribers subscribe specific news topics to the message broker, and the connection between the subscriber and publishers managed by the message broker. The data are transferring across the network when connecting to the message broker. When the message broker receives the published messages, it delivers the message to subscriber.The system designed also allows multiple users to subscribe messages with a subject because we need control several smart meter by groups, then the message broker will broadcast to different users.
- Broker: The Broker is the software in charge of getting the messages to the clients to which they have subscribed. The broker installed in our server is Mosca, it is an open-source MQTT broker that supports the MQTT V3.1 protocol and it is available for Linux [31]. In addition, it has a client running, responsible for subscribing the desired topics, receiving the data, decoding it (since it will be sent as a JSON string) and storing it appropriately.

A. MQTT over GPRS

The module used for GPRS communication in smart meters is a SIM 800 shield, this module supports the frequencies 850/900/1800/1900 MHz, i.e. the frequencies assigned to the mobile telecommunications operators in Ecuador.

We use the libraries available in the repositories of the RMAP group, to configure the SIM800 shield on the device (Arduino Mega) and modify the PubSubClient library for sending data on GPRS. The PubSubClient library has the features to publish, subscribe and LWT, but does not support GPRS. For the GPRS connection, the PubSubClient library requires a TCP connection through a Client.h file that generates TCP connections. To achieve what is described, we add in the PubSubClient library to ”include” command to define as TCP client and relate it as client of the SIM800 library [26].

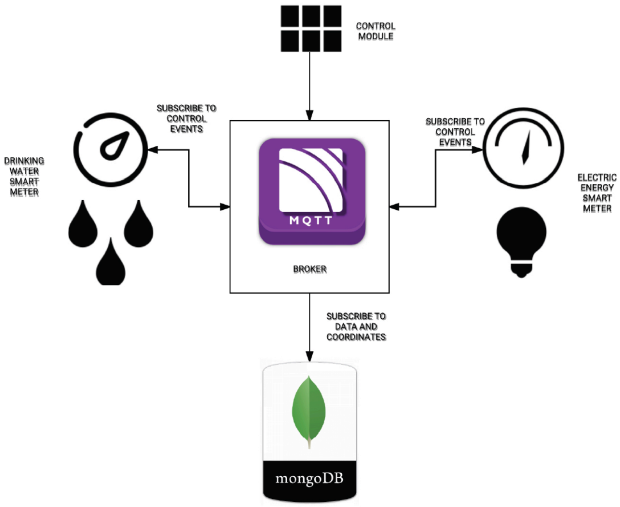


Figure 5: MQTT Broker

IV. Test and Results

A. Test

The system tests were performed as follows:

- Connectivity Test: Tests of connectivity between the intelligent sensors of potable water and electricity with the broker MQTT, using GPRS, using SIM Card of different mobile telecommunications operators.
- Bi-directional connectivity: For this test, we sent on / off messages from the MQTT corridor to Smart Meters and from the smart meters data was sent to the corridor MQTT.
- Data Base: To test the consistency of the information in the database, real measurements were sent from the meters built every 15 minutes, and a system was designed to simulate smart meters, in order to generate random data and test the scalability of the system.
- Management system tests:With the information received from the smart meters and the one generated by the system that simulates the generation of data, tests were performed to obtain individual, group and control statistics of the smart meters.

B. Results

The results of the connectivity tests performed with DATA SIM CARDs from three operators in covered locations did not present a problem. MQTT packets sent by smart meters were

not lost during a 15-minute period of about 96 messages per sensor.

The results of the tests on the database were carried out with the measurements sent by the intelligent meters and by the system of random generation of data for drinking water and electric energy arriving to have 1'465,279 registers, corresponding to 406 meters, this corresponds to 86 Records per day. The total size of the database was 351.62 MB. In the Fig. 6 shows the statistics of the project database.

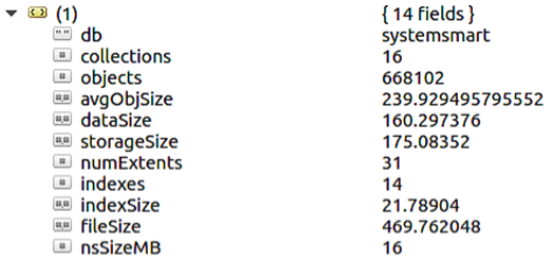


Figure 6: Test Results Data Base [26]

On the other had, the test results of the management and pre-billing system were performed by generating statistics by user, user groups and verification of the consistency of the data stored in the database, by means of a random verification of the information sent by the intelligent sensors. In the Fig. 7.

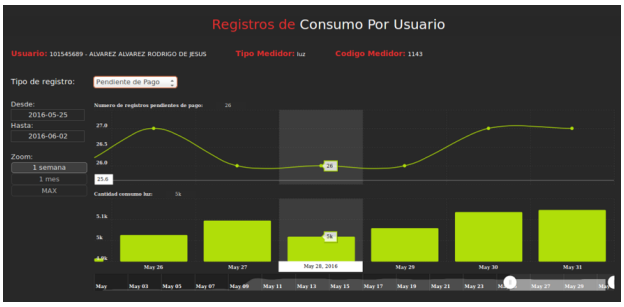


Figure 7: Test Results Statistics [26]

Likewise, the informatics system allows search the smart meters by number of customer or group of clients and showing the result over the city map. This shown in the Fig. 8.

V. Conclusions

Our design, building and testing of extensive work outlined in this paper, allowed to develop a informatics system for the registration of the measurement carried out smart meters of drinking water and electricity, as a functional contribution to the efficiency in the process of automated billing of these public services, through the use MQTT protocol over GPRS.

The use of mobile access networks to provide connectivity to IoT devices provides security in the transport of information and allows companies to have consistent data in real time. Likewise, centralized control of smart meters allows immediate action on the activation or deactivation of services.

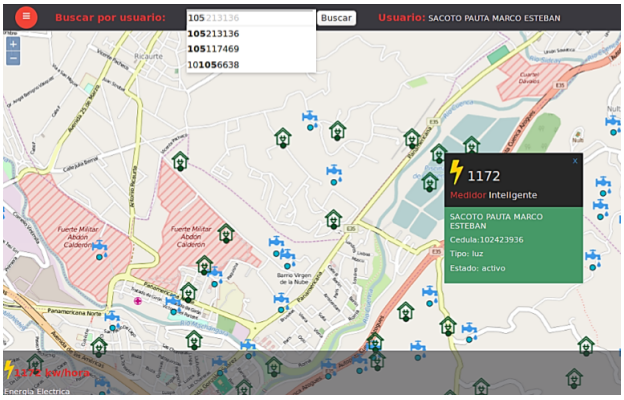


Figure 8: Test Results Smart Meter Location [26]

For the future works will be considered fog nodes by group of smart meters, and the Cloud application for management and control of smart devices.

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